NEUROPHYSIOLOGY MIDTERM, PRACTICE 1

Physiology, Year 2. Practice paper written by Sina Med from Prof. Covasa's lectures. One correct answer per question. 25 questions, 30 minutes.

1- After leaving the fourth ventricle, cerebrospinal fluid reaches the subarachnoid space mainly through

A- The cerebral aqueduct
B- The choroid plexus of the lateral ventricle
C- The arachnoid granulations
D- The foramen of Magendie
E- The foramen of Monro

2- Which sympathetic target receives postganglionic fibres that release acetylcholine?

A- The sinoatrial node
B- Bronchial smooth muscle
C- The dilator muscle of the pupil
D- Sweat glands
E- Renal blood vessels

3- A neurologist places a vibrating tuning fork on a patient's fingertip. The receptor mainly responsible for detecting the vibration is the

A- Free nerve ending
B- Pacinian corpuscle
C- Merkel disc
D- Ruffini ending
E- Meissner corpuscle

4- A man touches a hot pan. He first feels a sharp, well-localised pain, followed a second later by a dull burning pain. The burning second pain is carried by

A- Thinly myelinated A-delta fibres
B- A-alpha fibres from muscle spindles
C- Large myelinated A-beta fibres
D- Unmyelinated C fibres
E- Gamma motor fibres

5- The dorsal (posterior) column pathway carries

A- Visceral pain from the abdominal organs
B- Pain, temperature and crude touch
C- Automatic postural tone from the vestibular nuclei
D- Voluntary motor commands to distal limbs
E- Discriminative touch, vibration and conscious proprioception

6- Thanks to the buoyancy provided by cerebrospinal fluid, the effective weight of a 1400 g brain is reduced to about

A- 5 g
B- 700 g
C- 350 g
D- 50 g
E- 150 g

7- At rest, the membrane potential of a neuron (about -70 mV) lies closest to the equilibrium potential of

A- K^+, because K^+ leak channels are open
B- Na^+, because the Na^+/K^+ pump moves 3 Na^+ out
C- Ca^{2+}, because Ca^{2+} is high outside
D- Cl^-, because Cl^- is actively pumped in
E- Na^+, because Na^+ leak channels are open

8- The equilibrium potential of Na^+ in a typical neuron is approximately

A- -70 mV
B- +60 mV
C- -90 mV
D- -55 mV
E- 0 mV

9- A toxin removes the inactivation gate of voltage-gated Na^+ channels but leaves activation intact. The most likely effect on the action potential is

A- A larger afterhyperpolarization only
B- Loss of the rising phase
C- A faster repolarization
D- A more negative resting potential
E- A prolonged depolarization, because Na^+ keeps entering

10- Saltatory conduction is possible because voltage-gated channels are concentrated at the

A- Dendritic spines
B- Cell body
C- Nodes of Ranvier
D- Internodes beneath the myelin
E- Axon terminals only

11- In a typical multipolar neuron, the action potential is initiated at the

A- Axon terminal
B- Axon hillock
C- Node of Ranvier nearest the terminal
D- Cell body nucleus
E- Dendrites

12- Which statement correctly describes a graded potential?

A- It always depolarizes the membrane
B- It is all-or-none
C- Its amplitude is proportional to the stimulus and it decreases with distance
D- It travels long distances without decrement
E- It occurs only in membranes with voltage-gated channels

13- The afterhyperpolarization at the end of an action potential is mainly caused by

A- Closure of K^+ leak channels
B- Entry of Ca^{2+} through ligand-gated channels
C- Slow closure of voltage-gated K^+ channels
D- Activation of the Na^+/K^+ pump alone
E- Opening of voltage-gated Na^+ channels

14- A drug blocks voltage-gated Ca^{2+} channels in the presynaptic terminal. The action potential still reaches the terminal normally. What is the most likely result?

A- Neurotransmitter reuptake is increased
B- Acetylcholinesterase is inhibited
C- The action potential can no longer propagate along the axon
D- Postsynaptic receptors are blocked
E- Neurotransmitter release is reduced

15- The main inhibitory neurotransmitter in the brain is

A- GABA
B- Glutamate
C- Noradrenaline
D- Acetylcholine
E- Substance P

16- An inhibitory postsynaptic potential (IPSP) is typically produced by

A- Entry of Cl^-, which hyperpolarizes the membrane
B- Entry of Ca^{2+} into the presynaptic terminal
C- Closure of K^+ leak channels
D- Entry of Na^+, which depolarizes the membrane
E- Activation of voltage-gated Na^+ channels

17- A single presynaptic neuron fires three action potentials in rapid succession at the same synapse, and the postsynaptic neuron reaches threshold. This is an example of

A- Temporal summation
B- Habituation
C- Spatial summation
D- Divergence
E- Lateral inhibition

18- Selective serotonin reuptake inhibitors (SSRIs) act by

A- Keeping serotonin longer in the synaptic cleft
B- Converting serotonin into GABA
C- Blocking serotonin receptors
D- Increasing serotonin breakdown by enzymes
E- Preventing serotonin vesicle release

19- You no longer feel your watch on your wrist a few minutes after putting it on. This is mainly due to

A- Sensitization by serotonin
B- Rapid adaptation of phasic touch receptors
C- Damage to the dorsal column
D- Slow adaptation of tonic proprioceptors
E- Lateral inhibition in the thalamus

20- A patient has a hemisection of the right half of the spinal cord at T10. Which finding is expected below the lesion?

A- Loss of vibration sense on the left side
B- Loss of pain and temperature on the right side
C- Loss of all sensation on both sides
D- Loss of pain and temperature on the left side
E- Loss of fine touch on the left side

21- In smooth muscle, the Ca^{2+}-binding protein that starts contraction is

A- Titin
B- Calmodulin
C- Tropomyosin
D- Nebulin
E- Troponin

22- During contraction of a skeletal muscle fibre, which region keeps a constant length?

A- The H zone
B- The sarcomere
C- The A band
D- The I band
E- The distance between Z discs

23- A triad in skeletal muscle is made of

A- Three myofibrils bound by titin
B- Two T-tubules and one terminal cisterna
C- Actin, myosin and troponin
D- One Z disc and two M lines
E- One T-tubule and two terminal cisternae of the sarcoplasmic reticulum

24- A muscle tissue is striated, involuntary and its cells are linked by gap junctions. It is

A- Skeletal muscle
B- Multi-unit smooth muscle
C- Cardiac muscle
D- Myoepithelium
E- Single-unit smooth muscle

25- A student holds a heavy suitcase still with the elbow flexed at 90°. The biceps develops tension without changing length. This is

A- A passive stretch
B- A tetanic shortening
C- An isotonic eccentric contraction
D- An isotonic concentric contraction
E- An isometric contraction

Answer grid

Q	Answer	Q	Answer	Q	Answer	Q	Answer	Q	Answer
1		6		11		16		21	
2		7		12		17		22	
3		8		13		18		23	
4		9		14		19		24	
5		10		15		20		25	